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Project Title:	Oxidative Stress and Charged Particle Irradiation Alter Multipotent Stem Cells to Elicit Acute and Functional Changes in Target Organ Systems		
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	POSTDOCTORAL FELLOWSHIP Multipotent stem cells exist throughout the body and maintain the health of a given tissue by replenishing cells lost or damaged throughout life. These unique cells retain the capability to proliferate, migrate, and differentiate to return functionality to a tissue compromised by age, disease or a variety of geno/cytotoxic stressors such as ionizing radiation. Previously we showed that relatively low doses of radiation found in space (protons, heavy ions) can compromise the function of specific tissues and organs through a variety of biochemical mechanisms that involve oxidative stress. Oxidative stress that persists after irradiation can reduce bone density, impair muscle regrowth and inhibit neurogenesis. These acute, functional decrements could severely compromise the capability of astronauts to complete mission critical tasks, where relatively mild exposures to the space radiation environment may accelerate muscle fatigue and confusion		

Task Description:

caused by muscle and neural stem cell loss. These debilitating conditions can, in part, be caused by radiation-induced oxidative stress. Our data suggest that these multipotent cells may represent critical targets for countermeasures aimed at ameliorating radiation and space flight related sequelae. Thus, we propose in vitro and in vivo experiments to determine if and how relatively low doses of protons can initiate radioadaptive changes in stem cells found in the brain and skeletal muscle. Our proposal will evaluate how low dose proton exposure will impact stem cell proliferation, differentiation and survival and whether such endpoints can be altered by prior exposure. We hypothesize that stem cells incurring particle traversals during spaceflight can undergo adaptive changes that attenuate their response to subsequent larger exposures (e.g. from a solar flare). We will also investigate whether the mechanistic basis of adaptation involves changes in the redox environment such as oxidative stress. Three aims are proposed:

Specific Aim 1: To determine if/how low dose proton irradiation elicits radioadaptive changes in neural precursor cells from the CNS that depend on oxidative stress.

Specific Aim 2: To determine the acute dose response of satellite cells to proton irradiation with respect to survival, phenotypic fate and oxidative stress.

Specific Aim 3: Elucidate how low dose irradiation impacts neurogenesis in mice expressing human catalase targeted to the mitochondria (MCAT).

In aims 1 and 2, a range of biochemical, oxidative, and immunohistochemical approaches will be used that involve the quantification of reactive oxygen and nitrogen species and the detection of oxidative damage in neural precursor and satellite cells. In aim 3, we will utilize mice genetically modified to minimize mitochondrial hydrogen peroxide to determine how reducing intracellular oxidants impacts neurogenesis. These studies will help establish whether adaptation occurs after low dose proton irradiation, and whether stem cells and oxidative stress represent logical targets for countermeasure strategies.

The key findings are that low dose proton irradiation of rodent and human derived NPCs and rodent satellite cells results in ROS and RNS levels that fluctuate over time as well as a gradual increase in superoxide levels. Further, both dose and dose rate may be key components of radioadaptation in the multipotent cells when examining survival/proliferation using our priming-challenge paradigm.

The main impact of these finding is that we will focus on better understanding the radioadaptive effect of low dose and low dose rate proton irradiation on the NPCs and satellite cells.

The plan for the coming year is to determine the dose limits and dose rate dependence of the priming dose with respect to eliciting radioadaptation and the temporal interval during which the cells are radioadapted. Lastly, work will continue in determining whether or not the MCAT mice are radioprotected with respect to neurogenesis.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Very little is known about the low dose and low dose rate radiobiology of neural precursor cells. These studies will yield important information regarding how this radioresponse differs from the response to higher doses of ionizing radiation. Further, increased reactive oxygen species (ROS) can be damaging by challenging cellular repair and renewal processes but it can also activate redox sensitive signaling pathways which may underlie an adaptive response in neural precursors. Our recent work has begun to elucidate the effectiveness of low dose rate irradiation on the radioadaptation of neural precursor cells and it may prove to be quite beneficial as these dose rates better simulate the space radiation environment. Knowledge gained from understanding the mechanisms of change in oxidative stress, mitochondrial function and redox sensitive signaling in these experiments will provide the rationale for potentially effective countermeasure design for both acute and chronic space radiation exposure.
Ionizing radiation significantly reduces dentate neurogenesis, and such changes are dose dependent and persistent. These effects are linked to alterations in the neurogenic microenvironment, including inflammatory changes and oxidative stress although the precise mechanisms responsible are not yet known. While ROS have often been considered to be hostile or destructive entities they also have been shown to have beneficial effects, at least in part due to their role as signaling moieties. It has recently been shown that a persistent level of oxidative stress in extracellular superoxide dismutase knockout mice (EC-SOD KO) was associated with a lower baseline level of neurogenesis relative to wildtype (WT) mice. However, when those same mice were subjected to a modest dose of x-rays (5 Gy), there was no effect on neurogenesis in KO mice but a significant reduction in WT mice. Thus, we saw both negative (baseline neurogenesis) and positive (adaptive) effects in the EC-SOD KO mice, presumably as a result of mechanisms associated with redox balance. Also we have also been able to demonstrate the paradoxical nature of oxidative species in vitro using cultured neural precursor cells, where excess hydrogen peroxide reduces survival while excess superoxide increases survival. These paradoxical effects highlight the potential importance of adaptive responses in the context of the delicate balance in redox homeostasis, and how that may ultimately affect cell or tissue function. We believe that low doses of irradiation will elevate the level of oxidative stress in the neurogenic microenvironment and that this may have a beneficial effect when buffered by enhanced catalase activity with respect to dentate neurogenesis. These experiments can be conducted using a transgenic mouse with targeted expression of human catalase in the mitochondria (MCAT). Given the association between neurogenesis and cognitive function after irradiation, this modulation of oxidative stress via catalase could ultimately be developed into a countermeasure to maintain behavioral performance while engaged in space exploration.

With the capability to culture satellite cells in a multipotent state, we can now investigate the mechanistic details of ROS and RNS production in live cells. Changes in ROS and RNS following proton irradiation may prove be similar to what we have observed in either neural precursors exposed to the same irradiation at the same doses suggesting a common radioresponse pathway or quite different suggesting a more cell type specific response. Similarly mitochondrial redox function via superoxide levels can be readily assayed in these. Understanding these effects of irradiation on myogenic oxidative stress, proliferation and differentiation will help to provide the basis in order to design countermeasures to maintain muscle function during space exploration.

Task Progress:	Significant progress has been made over the last year. Work at Loma Linda University Medical Center has continued and the facility has been very accommodating to the doses, dose rates and irradiation paradigms that have been requested. The inability to perform the requested irradiations has decreased (although still present on more rare occasions) and overall we are very satisfied with this collaboration. With respect to the single dose experiments, ROS levels in neurospheres are perturbed by low doses of proton irradiation and may be dependent on dose rate. ROS, RNS and superoxide levels are elevated in hNSCs following proton irradiation and are less affected by dose rate. Thus far, mitochondrial content and function in hNSCs are not significantly changed at 60h following proton irradiation. Lastly, ROS, RNS and superoxide levels are elevated in satellite cells following (high dose rate) HDR proton irradiation, with the LDR (low dose rate) studies pending. With regard to the priming-challenge irradiation paradigm, a 25 cGy low dose (both HDR and LDR) proton priming dose improves 5 day survival in hNSC following a 1 Gy but not a 2 Gy gamma challenge dose 24h post priming whereas 25 cGy HDR but not LDR priming improves satellite cell survival after 1 and 2 Gy gamma challenge doses 24h post priming. These data suggest that priming doses may depend on dose and dose rate in different precursor cells types. Finally, the MCAT have been irradiated with 0.5 and 2 Gy of proton irradiation. The mice were injected with BrdU for neurogenesis studies and the brains have been harvested. A majority of the brains have been sectioned. Some technical difficulties were encountered with respect to immunohistochemically double labeling mouse tissue for neurogenic markers and BrdU using combinations of mouse and rat primary antibodies but these issues have been resolved and the immunohistochemical analysis for neurogenesis in underway.
Bibliography Type:	Description: (Last Updated: 08/12/2013)
Articles in Peer-reviewed Journals	Acharya MM, Lan ML, Kan VH, Patel NH, Giedzinski E, Tseng BP, Limoli CL. "Consequences of ionizing radiation-induced damage in human neural stem cells." Free Radic Biol Med. 2010 Sep 6. [Epub ahead of print]. PMID: 20826207 , Sep-2010
Articles in Peer-reviewed Journals	Caiozzo VJ, Giedzinski E, Baker M, Suarez T, Izadi A, Lan M, Cho-Lim J, Tseng BP, Limoli CL. "The radiosensitivity of satellite cells: cell cycle regulation, apoptosis and oxidative stress." Radiat Res. 2010 Nov;174(5):582-9. Epub 2010 Aug 2. PMID: 20726709 , Nov-2010